C++ Notes: Switch Statements

(Convert day number to day name)

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Learning Objectives

After this section, you will be able to:

- Use `switch` to compare one value against many cases
- Understand `case`, `break`, and `default`
- Build `getDayOfWeek(int)` that returns a `string`

Core Idea

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A `switch` is useful when you repeatedly compare the **same variable** against many values (0,1,2,3,...).
It is cleaner than a long chain of `if / else if`.

Function: getDayOfWeek

Listing 1: Switch example: day number to day name

```
#include <iostream>
#include <string>
using namespace std;

string getDayOfWeek(int dayNum) {
  string dayName;

  switch (dayNum) {
    case 0: dayName = "Sunday"; break;
    case 1: dayName = "Monday"; break;
    case 2: dayName = "Tuesday"; break;
    case 3: dayName = "Wednesday"; break;
    case 4: dayName = "Thursday"; break;
    case 5: dayName = "Friday"; break;
    case 6: dayName = "Saturday"; break;
    default: dayName = "Invalid day number";
  }

  return dayName;
}
```

```
int main() {
  cout << getDayOfWeek(0) << endl;
  cout << getDayOfWeek(1) << endl;
  cout << getDayOfWeek(6) << endl;
  cout << getDayOfWeek(10) << endl;
  return 0;
}
```

Output

```
Sunday
Monday
Saturday
Invalid day number
```

Why we need `break`

Without `break`, execution "falls through" into the next case, and can overwrite your result.

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